

LV Dimensions



2D-guided linear measurements

LV internal dimension	Mean $\pm$ SD	2-SD Range	Mean $\pm$ SD	2-SD Range
Diastolic dimension (mm)	50.2 $\pm$ 4.1	42.0 - 58.4	45.0 $\pm$ 3.6	37.8 - 52.2
Systolic dimension (mm)	32.4 $\pm$ 3.7	25.0 - 39.8	28.2 $\pm$ 3.3	21.6 - 34.8

LV Volumes by 3D



Volumetric measurements

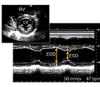
Upper limits of normal

End-diastolic volume:
79 mL/m² for men
71 mL/m² for women

End-systolic volume:
32 mL/m² for men
28 mL/m² for women

LV Mass

Linear method



Cube formula
 $LV\ mass = 0.8 \times 1.04 \times \frac{3}{4} \pi [(LV\ intd + HW)^2 \times LV\ intd] \times 0.6$

	Women	Men
LV mass/BSA (g/m ²)	43-65	49-71.5
Relative wall thickness (cm)	0.22-0.42	0.24-0.42
Septal thickness (cm)	0.6-0.9	0.6-1.0
Posterior wall thickness (cm)	0.6-0.9	0.6-1.0

BSA: body surface area; recommended and best selected.

2D Methods



Two methods for estimating LV mass based on area-length formula and the truncated ellipsoid formula, from short axis (top) and apical four-chamber (bottom) 2D echo views.

	Women	Men
LV mass/BSA (g/m ²)	44-65	50-102

BSA: body surface area; recommended and best selected.

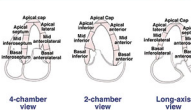
Relative Wall Thickness ≥ 1.0	Concentric Remodeling	Concentric Hypertrophy
Relative Wall Thickness < 1.0	Normal Geometry	Eccentric Hypertrophy

Left Ventricular Mass Index (g/m²)

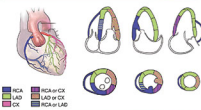
< 95 (1)
 > 115 (1)

> 95 (1)
 > 115 (1)

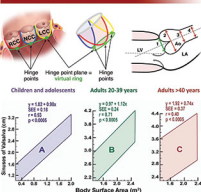
LV Segmentation



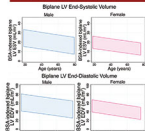
Perfusion Territories



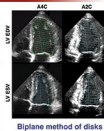
Aortic Root



LV Volumes by 2D



LV volumes normalized by BSA	Mean $\pm$ SD	2-SD Range	Mean $\pm$ SD	2-SD Range
LV end-diastolic volume (mL/m ²)	54 $\pm$ 10	34 - 74	45 $\pm$ 8	29 - 61
LV end-systolic volume (mL/m ²)	21 $\pm$ 5	11 - 31	16 $\pm$ 4	8 - 24



LV Function

Ejection fraction

	Normal range	Mildly abnormal	Moderately abnormal	Severely abnormal
Male	52 - 72	41 - 51	30 - 40	< 30
Female	54 - 74	41 - 53	30 - 40	< 30

Global longitudinal strain

Peak GLS in the range of -20% can be expected in a healthy person, and the lower the absolute value of strain is, the more likely it is to be abnormal.



LA Volume

Biplane method of disks

Biplane method of disks, using the apical four-chamber (A4C) and apical two-chamber (A2C) views at end-diastole and end-systole (maximum LA size).

Area-length technique

$$\frac{8}{3\pi} \left[\frac{A_1 \times A_2}{L} \right]$$

Area-length method using the apical four-chamber and apical two-chamber views at end-diastole and end-systole (maximum LA size). The length (L) is measured from the back wall to the line across the hinge points of the mitral valve. The shorter L from either the A4C or the A2C is used in the equation.

	Normal range	Mildly abnormal	Moderately abnormal	Severely abnormal
Maximum LA volume / BSA (mL/m ²)	16 - 34	35 - 41	42 - 48	> 48